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Coagulation mold, a kit and a method for preparing a coagulated blood mass

Abstract

The present disclosure provides a method, a coagulation mold and a kit for preparing a coagulated blood mass having a desired shape and volume. The coagulated blood mass is prepared from whole blood being withdrawn from a subject. The whole blood is mixed with coagulation initiators in order to obtain the coagulated blood mass with optimal parameters. This is obtained, inter alia, by selecting particle size of one of the coagulation initiators such as to yield more rapid dissolving within the blood and/or to yield improved suspension of the particles in the blood when it is still in a liquid form. By controlling these parameters, an optimal coagulated blood mass is obtained that can be used for treatment of the human body, e.g. for treating skin lesions, internal injuries, such as anal fistula, stoma cavity, damaged tendon, and others.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of International Application No. PCT/IL2023/050083 filed Jan. 25, 2023, which claims priority to Israel patent application number 290122 filed Jan. 25, 2022, the disclosures of which are incorporated herein by reference in their entirety.

TECHNOLOGICAL FIELD

(1) The present disclosure is in the field of preparation of medical treatment, in particular preparation of a coagulated blood mass for use in a treatment of a subject.

BACKGROUND ART

(2) References considered to be relevant as background to the presently disclosed subject matter are listed below: U.S. Pat. No. 9,180,142 US 2020/0281775 WO 2019/150355

(3) Acknowledgement of the above references herein is not to be inferred as meaning that these are in any way relevant to the patentability of the presently disclosed subject matter.

GENERAL DESCRIPTION

(4) The present disclosure provides a method, a coagulation mold and a kit for preparing a coagulated blood mass, namely blood clot having a desired shape and volume. The coagulated blood mass is prepared from whole blood being withdrawn from a subject. The whole blood is mixed with coagulation initiators, i.e. coagulation agents or anti-anti coagulation agents, in a specific amount and having specific characteristics in order to obtain the coagulated blood mass with optimal parameters, such as stability, moisture, homogeneousness, namely the texture of the clot is substantially similar at any portion thereof, etc. This is obtained, inter alia, by selecting particle size of one of the coagulation initiators such as to yield more rapid dissolving within the blood and/or to yield improved suspension of the particles in the blood when it is still in a liquid form. Another parameter to yield the desired coagulated blood mass of the present invention is the selected amount of material of coagulation initiator to be mixed with the whole blood. By controlling these parameters, an optimal coagulated blood mass is obtained that can be used for treatment of the human body, e.g. for treating skin lesions, internal injuries (such as anal fistula, stoma cavity, damaged tendon, etc.).

(5) Thus, a first aspect of the present disclosure provides a method for preparing a coagulated blood mass, i.e. a volumetric blood clot, for use in a treatment of a lesion or injury of a subject. The treatment can be on the external portion of the body such as skin lesions or on internal portions of the body such as fistula cavity, abdominal cavity, etc. The method comprising: mixing an amount of whole blood with calcium gluconate powder and kaolin powder; and allowing the whole blood to coagulate to form the coagulated blood mass. The term “allowing” defines that the mixture of blood with maintaining the mixture of whole blood with calcium gluconate powder and kaolin powder is maintained for a selected amount of time until the blood solidifies to a certain degree and seize to be in a flowable form. The amount of time can be more than 1 minute, 2, 3, 4, 5, 6, 7, 8, 9, 10 or even more than 15 minutes until the whole blood is solidified to the desired degree. The method is further characterized by at least one of: (i) the particle-size distribution D50 of the calcium gluconate powder is between 50 μ -200 μ , (ii) wherein said mixing comprises mixing

between 0.6 mg-3.1 mg of kaolin powder with every 1 ml of whole blood, (iii) wherein said mixing comprises mixing between 2.5 mg-8.1 mg of calcium gluconate powder with every 1 ml of whole blood, or (iv) the particle-size distribution D50 of the kaolin powder is less than 20 μ .

(6) The particle-size distribution DX means that X % of the particles in the powder are less than the nominal size associated with said distribution. For example, if the particle-size distribution D50 is 125 μ then 50% of the particles are having a volumetric size equal or less to 125 μ . By maintaining only particles with size that is relatively small, the resulted coagulated blood mass is homogenous, namely the volumetric solidity of the coagulated blood mass is substantially equal. This is resulted due to the duration that particles with such a size can suspend in the whole blood they are mixed with. Larger particles tend to sink in the blood and the resulted coagulated blood mass is not homogeneous.

(7) It is to be noted that any combination of the described embodiments with respect to any aspect of this present disclosure is applicable. In other words, any aspect of the present disclosure can be defined by any combination of the described embodiments.

(8) In some embodiments of the method, the particle-size distribution D70 of the calcium gluconate powder is between 50 μ -200 μ .

(9) In order to prepare a coagulated blood mass with properties that are suitable for using in a medical treatment the calcium gluconate powder is first milled to a size of 2 μ . The particles product of this milling is typically undergoing a certain degree of granulation, thus prior to the use of the powder in the formation of the coagulated blood mass, the calcium gluconate powder is sifted through a sifter of 120-125 μ particles. Particle size of less than 120-125 μ dissolve much faster and therefore the use thereof in the formation of a coagulated blood mass results in a much more stable and non-leaky coagulated blood mass.

(10) In some embodiments of the method, particle-size distribution D90 of the calcium gluconate powder is between 50 μ -200 μ .

(11) In some embodiments of the method, the particle-size distribution D90, D80, D70, D60, or D50 of the calcium gluconate powder is about 125 μ .

(12) The term "about" should be understood as a deviation of up to 20% from the nominal value. For example, if the nominal value is about 125 μ , then the value should be considered as 100 μ -150 μ .

(13) In some embodiments of the method, the particle-size distribution of either D50, D60, D70, D80, D90 or D95 of the calcium gluconate powder is between either 50 μ -200 μ , 60 μ -190 μ , 70 μ -180 μ , 80 μ -170 μ , 90 μ -160 μ , 100 μ -150 μ , 110 μ -140 μ or 120 μ -130 μ .

(14) In some embodiments of the method, the particle-size distribution D99 of the kaolin powder is less than 50 μ .

(15) In some embodiments of the method, the particle-size distribution D99 of the kaolin powder is less than 50 μ .

(16) By using small particles of kaolin in the preparation of the coagulated blood mass, the kaolin particles suspend for a longer duration in the formed coagulating blood mass (when at least some of the blood is in liquid flowable form) and do not sink rapidly to the bottom of the formed coagulating blood mass, therefore resulting in a substantive homogeneous coagulated blood mass.

(17) In some embodiments of the method, the particle-size distribution D80 of the kaolin powder is less than 25 μ .

(18) In some embodiments of the method, the particle-size distribution D90 of the kaolin powder is less than 3 μ or between 1 μ -3 μ .

(19) In some embodiments of the method, the particle-size distribution of either D50, D60, D70, D80, D90 or D95 of the calcium gluconate powder is less than either 30 μ , 25 μ , 20 μ , 15 μ , 10 μ , 5 μ , 3 μ , 2 μ , 1 μ or less than 0.1 μ .

(20) In some embodiments, the method comprising withdrawing said whole blood from the subject, therefore the coagulated blood mass is prepared from autologous whole blood.

- (21) In some embodiments, the method comprising, prior to said mixing, introducing anti-coagulant-containing liquid to said amount of whole blood. Namely, after the whole blood is withdrawn, it is added with anti-coagulant to prevent undesired coagulation of the blood.
- (22) In some embodiments of the method, said introducing comprises introducing between 0.04 ml-0.095 ml of anti-coagulant-containing liquid for every 1 ml of whole blood.
- (23) In some embodiments of the method, said introducing comprises introducing about 0.07 ml of anti-coagulant-containing liquid for every 1 ml of whole blood.
- (24) In some embodiments of the method, said mixing is performed within a coagulation mold. The coagulation mold has a desired shape that determines the shape of the blood clot being prepared by the method.
- (25) In some embodiments of the method, the mixing comprises mixing between 0.6 mg-3.1 mg of kaolin powder with every 1 ml of whole blood.
- (26) In some embodiments of the method, said mixing comprises mixing about 1.75 mg of kaolin powder with every 1 ml of whole blood.
- (27) In some embodiments of the method, said mixing comprises mixing between 0.8 mg-2.9 mg, 1 mg-2.7 mg, 1.2 mg-2.5 mg, 1.4 mg-2.3 mg, 1.6 mg-2.1 mg or 1.8 mg-2 mg of kaolin powder with every 1 ml of whole blood.
- (28) In some embodiments of the method, said mixing comprises mixing between 2.5 mg-8.1 mg of calcium gluconate powder with every 1 ml of whole blood.
- (29) In some embodiments of the method, said mixing comprises mixing about 5.3 mg of calcium gluconate powder with every 1 ml of whole blood.
- (30) In some embodiments of the method, said mixing comprises mixing between 2.9 mg-7.7 mg, 3.3 mg-7.3 mg, 3.7 mg-6.9 mg, 4.1 mg-6.5 mg, 4.5 mg-6.1 mg or 4.9 mg-5.7 mg of calcium gluconate powder with every 1 ml of whole blood.
- (31) Yet another aspect of the present disclosure provides a coagulation mold comprising a coagulation space defined by a base surrounded by upwardly rising walls and configured to receive a selected amount of whole blood. The coagulation space comprises calcium gluconate powder and kaolin powder. The coagulation mold comprises at least one of the following: (i) the particle-size distribution D50 of the calcium gluconate powder between 50 μ -200 μ , (ii) kaolin powder in an amount of between 0.6 mg-3.1 mg for every 1 ml of said selected amount of whole blood, (iii) calcium gluconate in an amount of between 2.5 mg-8.1 mg for every 1 ml of said selected amount of whole blood, or (iv) the particle-size distribution D50 of the kaolin powder is less than 10 μ .
- (32) In some embodiments of the coagulation mold, the particle-size distribution D70 of the calcium gluconate powder is between 50 μ -200 μ .
- (33) In some embodiments of the coagulation mold, the particle-size distribution D90 of the calcium gluconate powder is between 500 μ -200 μ .
- (34) In some embodiments of the coagulation mold, the particle-size distribution D90, D80, D70, D60, or D50 of the calcium gluconate powder is about 125 μ .
- (35) In some embodiments of the coagulation mold, the particle-size distribution of either D50, D60, D70, D80, D90 or D95 of the calcium gluconate powder is between either 50 μ -200 μ , 60 μ -190 μ , 70 μ -180 μ , 80 μ -170 μ , 90 μ -160 μ , 100 μ -150 μ , 110 μ -140 μ or 120 μ -130 μ .
- (36) In some embodiments of the coagulation mold, the particle-size distribution D99 of the kaolin powder is less than 50 μ .
- (37) In some embodiments of the coagulation mold, the particle-size distribution D90 of the kaolin powder is less than 20 μ or less than 10 μ .
- (38) In some embodiments of the coagulation mold, the particle-size distribution D50 of the kaolin powder is less than 3 μ or between 1 μ -3 μ .
- (39) In some embodiments of the coagulation mold, the particle-size distribution of either D50, D60, D70, D80, D90 or D95 of the calcium gluconate powder is less than either 30 μ , 25 μ , 20 μ , 15 μ , 10 μ , 5 μ , 3 μ , 2 μ , 1 μ or less than 0.1 μ .

- (40) In some embodiments of the coagulation mold, the whole blood being intended to be received therein comprises between 0.04 ml-0.095 ml of anti-coagulant-containing liquid for every 1 ml of whole blood or in some embodiments about 0.07 ml of anti-coagulant-containing liquid for every 1 ml of whole blood.
- (41) In some embodiments, the coagulation mold comprising between 0.6 mg-3.1 mg of kaolin powder for every 1 ml of said selected amount of whole blood.
- (42) In some embodiments, the coagulation mold comprising about 1.75 mg of kaolin powder for every 1 ml of said selected amount of whole blood.
- (43) In some embodiments, the coagulation mold comprising between 0.8 mg-2.9 mg, 1 mg-2.7 mg, 1.2 mg-2.5 mg, 1.4 mg-2.3 mg, 1.6 mg-2.1 mg or 1.8 mg-2 mg of kaolin powder with every 1 ml of whole blood.
- (44) In some embodiments, the mold comprising between 2.5 mg-8.1 mg of calcium gluconate powder for every 1 ml of said selected amount of whole blood.
- (45) In some embodiments, the mold comprising about 5.3 mg of calcium gluconate powder for every 1 ml of said selected amount of whole blood.
- (46) In some embodiments, the coagulation mold comprising 2.9 mg-7.7 mg, 3.3 mg-7.3 mg, 3.7 mg-6.9 mg, 4.1 mg-6.5 mg, 4.5 mg-6.1 mg or 4.9 mg-5.7 mg of calcium gluconate powder with every 1 ml of whole blood.
- (47) In some embodiments of coagulation mold, the volume of the coagulation space is sufficient to receive between 13 ml-19 ml of whole blood.
- (48) In some embodiments of coagulation mold, the volume of the coagulation space is sufficient to receive about 16 ml of whole blood.
- (49) Yet another aspect of the present disclosure provides a kit for preparing a coagulated blood mass comprising: calcium gluconate powder; kaolin powder; and a coagulation mold for mixing a selected amount of blood with the calcium gluconate powder and kaolin powder. The calcium gluconate powder and the kaolin powder are characterized by at least one of the following: (i) the particle-size distribution D50 of the calcium gluconate powder is between 50 μ -200 μ , (ii) wherein said kaolin powder is an amount of between 0.6 mg-3.1 mg for every 1 ml of said selected amount of whole blood, (iii) wherein said calcium gluconate is an amount of between 2.5 mg-8.1 mg for every 1 ml of said selected amount of whole blood, or (iv) the particle-size distribution D50 of the kaolin powder is less than 10 μ .
- (50) In some embodiments, the kit comprising tools for withdrawing whole blood and containing it.
- (51) In some embodiments, the kit comprising 0.04 ml-0.095 ml of anti-coagulant-containing liquid for every 1 ml of said selected amount of whole blood. The anti-coagulant-containing liquid can be contained separately in the kit or within the tool for containing the withdrawn whole blood.
- (52) In some embodiments of the kit, said selected amount is between 13 ml-19 ml.
- (53) In some embodiments of the kit, the particle-size distribution D70 of the calcium gluconate powder is between 50 μ -200 μ .
- (54) In some embodiments of the kit, the particle-size distribution D90 of the calcium gluconate powder is between 50 μ -200 μ .
- (55) In some embodiments of the kit, the particle-size distribution D90, D80, D70, D60, or D50 of the calcium gluconate powder is about 125 μ .
- (56) In some embodiments of the kit, the particle-size distribution of either D50, D60, D70, D80, D90 or D95 of the calcium gluconate powder is between either 50 μ -200 μ , 60 μ -190 μ , 70 μ -180 μ , 80 μ -170 μ , 90 μ -160 μ , 100 μ -150 μ , 110 μ -140 μ or 120 μ -130 μ .
- (57) In some embodiments of the kit, the particle-size distribution D99 of the kaolin powder is less than 50 μ .
- (58) In some embodiments of the kit, the particle-size distribution D80 of the kaolin powder is less than 10 μ .
- (59) In some embodiments of the kit, the particle-size distribution D50 of the kaolin powder is less

than 3 μ or between 1 μ -3 μ .

(60) In some embodiments of the kit, the particle-size distribution of either D50, D60, D70, D80, D90 or D95 of the calcium gluconate powder is less than either 30 μ , 25 μ , 20 μ , 15 μ , 10 μ , 5 μ , 3 μ , 2 μ , 1 μ or less than 0.1 μ .

(61) In some embodiments, the kit comprising between 0.6 mg-3.1 mg of kaolin powder for every 1 ml of said selected amount of whole blood.

(62) In some embodiments, the kit comprising about 1.75 mg of kaolin powder for every 1 ml of said selected amount of whole blood.

(63) In some embodiments, the kit comprising between 0.8 mg-2.9 mg, 1 mg-2.7 mg, 1.2 mg-2.5 mg, 1.4 mg-2.3 mg, 1.6 mg-2.1 mg or 1.8 mg-2 mg of kaolin powder with every 1 ml of whole blood.

(64) In some embodiments, the kit comprising between 2.5 mg-8.1 mg of calcium gluconate powder for every 1 ml of said selected amount of whole blood.

(65) In some embodiments, the kit comprising about 5.3 mg of calcium gluconate powder for every 1 ml of said selected amount of whole blood.

(66) In some embodiments, the kit comprising 2.9 mg-7.7 mg, 3.3 mg-7.3 mg, 3.7 mg-6.9 mg, 4.1 mg-6.5 mg, 4.5 mg-6.1 mg or 4.9 mg-5.7 mg of calcium gluconate powder with every 1 ml of whole blood.

(67) In some embodiments, the kit comprising the coagulation mold of any one of the above-described embodiments or any combination thereof. The kaolin powder and calcium gluconate powder are comprised within said coagulation mold.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) In order to better understand the subject matter that is disclosed herein and to exemplify how it may be carried out in practice, embodiments will now be described, by way of non-limiting example only, with reference to the accompanying drawings, in which:

(2) FIGS. 1A-1B are images of the results of the kaolin powder particle size test. FIG. 1A shows the result in time zero and FIG. 1B shows the result in time 15 minutes.

DETAILED DESCRIPTION

(3) The following are examples of tests with varying parameters of the powders used in the method for forming a coagulated blood mass. These tests are provided to exemplify embodiments and realization of the invention of the present disclosure.

(4) Calcium Gluconate (CG) Powder Test

(5) Success Criteria:

(6) Optimal Clot is Defined by:

(7) Coagulation time—good coagulation time is under 6 minutes Visual clot structure strength—good structure strength is defined as the clot does not break down when extracted and handled by user CG residuals—visual assessment of amount of white powder residuals accumulated on the bottom of the clot—0=no residuals, 5=100% coverage of surface with residuals

Method:

A standard amount of ACD-A anticoagulant and Kaoilin powder per 1 ml of whole blood was used
2 ml of ACD-A for 15 ml blood=0.133 ml ACD-A/1 ml blood 28 mg Kaolin powder for 15 ml blood=1.87 mg Kaolin/1 ml blood

(8) Different amounts of CG where tested-CG particle size after sifting is 125 micron 20 mg for 15 ml of blood=1.33 mgCG/1 ml blood 50 mg for 15 ml of blood=3.33 mgCG/1 ml blood 85 mg for 15 ml of blood=5.66 mgCG/1 ml blood 110 mg for 15 ml of blood=7.33 mgCG/1 ml blood 140 mg for 15 ml of blood=9.33 mgCG/1 ml blood

(9) Coagulation time and clot structure strength were documented. Blood samples of 5 different volunteers were used.

(10) Results:

(11) The following table describes the results of the test:

(12) TABLE-US-00001 TABLE 1 Coagulation Clot Powder # volunteer/CG time strength residuals
#1/20 mg 15 Low 0 #1/50 mg 6.8 Medium 0 #1/85 mg 4.7 Strong 0 #1/110 mg 5 Strong 2 #1/140 mg 4.9 Strong 5 #2/20 mg 17 Low 0 #2/50 mg 6.7 Medium 0 #2/85 mg 4.8 Strong 0 #2/110 mg 4.7 Strong 2 #2/140 mg 4.8 Strong 5 #3/20 mg 14 Low 0 #3/50 mg 5.4 Medium 0 #3/85 mg 4.6 Strong 1 #3/110 mg 5.2 Strong 2 #3/140 mg 4.7 Strong 5 #4/20 mg 12 Low 0 #4/50 mg 6 Medium 0 #4/85 mg 4.8 Strong 0 #4/110 mg 4.6 Strong 2 #4/140 mg 5 Strong 5 #5/20 mg 14.5 Low 0 #5/50 mg 6.6 Medium 0 #5/85 mg 4.6 Strong 0 #5/110 mg 4.9 Strong 2 #5/140 mg 5.2 Strong 5 Average/20 mg 14.5 Low 0 Average/50 mg 6.3 Medium 0 Average/85 mg 4.88 Strong 0.2 Average/110 mg 4.7 Strong 2 Average/140 mg 4.92 Strong 5

Kaolin Powder Test

Success Criteria:

Optimal Clot is Defined by: 1. Coagulation time-good coagulation time is under 6 minutes 2.

Visual clot structure strength-good structure strength is defined as the clot does not break down when extracted and handled by user

Method:

A standard amount of ACD-A anticoagulant and Calcium Gluconate powder per 1 ml of whole blood was used 2 ml of ACD-A for 15 ml blood=0.133 ml ACD-A/1 ml blood 85 mg CG powder for 15 ml blood=5.66 mg CG/1 ml blood

Different amounts of Kaolin where tested-Kaolin particle size is 1-3 micron 5 mg for 15 ml of blood=1.33 mgCG/1 ml blood 10 mg for 15 ml of blood=3.33 mgCG/1 ml blood 20 mg for 15 ml of blood=5.66 mgCG/1 ml blood 30 mg for 15 ml of blood=7.33 mgCG/1 ml blood 40 mg for 15 ml of blood=9.33 mgCG/1 ml blood

Coagulation time and clot structure strength were documented. Blood samples of 5 different volunteers were used.

Results:

The following table describes the results of the test:

(13) TABLE-US-00002 TABLE 2 # volunteer/Kaolin Coagulation time Clot strength #1/5 mg 12 Low #1/10 mg 8.2 Medium #1/20 mg 6.2 Strong #1/30 mg 4.7 Strong #1/40 mg 4.9 Strong #2/5 mg 14 Low #2/10 mg 8.5 Medium #2/20 mg 5.8 Strong #2/30 mg 5.1 Strong #2/40 mg 5.5 Strong #3/5 mg 11.5 Low #3/10 mg 7.7 Medium #3/20 mg 5.4 Strong #3/30 mg 5.2 Strong #3/40 mg 4.8 Strong #4/5 mg 13.4 Low #4/10 mg 8.5 Medium #4/20 mg 6.3 Strong #4/30 mg 4.9 Strong #4/40 mg 5.2 Strong #5/5 mg 14 Low #5/10 mg 7.3 Medium #5/20 mg 5.4 Strong #5/30 mg 5.1 Strong #5/40 mg 5.1 Strong Average/5 mg 12.98 Low Average/10 mg 8.04 Medium Average/20 mg 5.82 Strong Average/30 mg 5 Strong Average/40 mg 5.1 Strong

Kaolin Powder Particle Size Test

Goal

(14) The goal of the test is to establish effective threshold of Kaolin powder to float in water for sufficient time to allow coagulation of blood simultaneously in all volume at the same time.

(15) Background

(16) Blood coagulates rapidly via the extrinsic pathway when Kaolin or other coagulation initiator (such as glass) comes in contact with factor XI. The more Kaolin particles that come in contact with blood for more time, the higher probability of contact. Kaolin is a clay mineral that typically sink in water due to its weight. If Kaolin particles will be able to freely float in liquid it will have higher effect of coagulation, achieving an evenly distributed coagulation that starts at once in all the blood volume, similar in the upper part of the blood volume as in the lower part of the blood volume etc. The idea is that if kaolin particles are sufficiently small, they can float in the blood for

a sufficient duration before they start to sink to allow homogenic coagulation initiation and acceleration, achieving a homogenic clot.

(17) Success Criteria

(18) Kaolin particles float in liquid for 15 minutes without sinking

Method 2 sizes of particles were tested Smaller than 25 micron Larger than 25 micron

(19) For each group of particles size, 50 mg of powder was placed inside a clear glass vessel with 160 ml of purified water. Both glass vessels were mixed for 5 seconds with a steering spoon. At time points 0 and 15 minutes the vessels were visually assessed for the following parameters 1. Water color and transparency (from 10: white/non transparent to 1: Clear/Fully Transparent) 2. Accumulation of kaolin powder on the bottom of the vessel (From 10: No accumulation to 1: fully accumulated)

Results

(20) The following table describes the results of the test (and also can be seen in FIGS. 1A-1B):

(21) TABLE-US-00003 TABLE 3 Water Accumulation on # color&transparency the bottom Time0

10 10 Cup <25 micron Time0 10 10 Cup >25 micron Time15 min 10 2 Cup <25 micron Time15 min 10 4 Cup <25 micron

CONCLUSIONS

(22) Kaolin particle size lower than 25 micron float in water and thus in blood for 15 minutes without sinking while kaolin particle larger than 25 micron will sink within timeframe of 15 minutes.

Claims

1. A coagulation mold comprising: a coagulation space defined by a base surrounded by upwardly rising walls and configured to receive a selected amount of whole blood; and calcium gluconate powder and kaolin powder comprised within said coagulation space; the coagulation mold being characterized in that: (i) said calcium gluconate powder is in an amount of about 3.3 mg-7.3 mg per 1 ml of said selected amount of whole blood; (ii) the particle-size distribution D50 of the calcium gluconate powder is about 50 μ -200 μ ; (iii) said kaolin powder is in an amount of about 0.6 mg-3.1 mg per 1 ml of said selected amount of whole blood; and (iv) the particle-size distribution D50 of the kaolin powder is less than about 25 μ .

2. The coagulation mold of claim 1, wherein the particle-size distribution of the calcium gluconate powder meets one of the criteria selected from (i) D70 of between about 50 μ -200 μ , (ii) D90 of between about 50 μ -200 μ , and (iii) D50 of about 125 μ .

3. The coagulation mold of claim 1, wherein the particle-size distribution of the kaolin powder meets one of the criteria selected from (i) D99 of less than about 50 μ , (ii) D80 of less than about 25 μ , and (iii) D45 of less than about 3 μ .

4. The coagulation mold of claim 1, comprising at least one of (i) about 1.75 mg of kaolin powder per 1 ml of whole blood, and (ii) about 5.3 mg of calcium gluconate powder per 1 ml of whole blood.

5. The coagulation mold of claim 1, wherein the volume of the coagulation space is sufficient to receive an amount of 13 ml-19 ml of whole blood.

6. A kit for preparing a coagulated blood mass comprising: calcium gluconate powder; kaolin powder; and a coagulation mold for mixing a selected amount of whole blood with the calcium gluconate powder and kaolin powder; the kit being characterized in that: (i) said calcium gluconate powder is in an amount of about 3.3 mg-7.3 mg per 1 ml of said selected amount of whole blood; (ii) the particle-size distribution D50 of the calcium gluconate powder is about 50 μ -200 μ ; (iii) said kaolin powder is in an amount of about 0.6 mg-3.1 mg per 1 ml of said selected amount of whole blood; and (iv) the particle-size distribution D50 of the kaolin powder is less than about 25 μ .

7. The kit of claim 6, further comprising tools for withdrawing whole blood and containing it.

8. The kit of claim 6, comprising about 0.04 ml-0.095 ml of anti-coagulant-containing liquid for every 1 ml of said selected amount of whole blood.

9. The kit of claim 6, wherein the particle-size distribution of the calcium gluconate powder meets one of the criteria selected from (i) D70 is between 50 μ -200 μ , (ii) D90 is between 50 μ -200 μ , and (iii) D50 is about 125 μ .

10. The kit of claim 6, wherein the particle-size distribution of the kaolin powder meets one of the criteria selected from (i) D99 is less than about 50 μ , (ii) D80 is less than about 25 μ , and (iii) D45 is less than about 3 μ .

11. The kit of claim 6, comprising at least one of (i) about 1.75 mg of kaolin powder per 1 ml of whole blood, and (ii) about 5.3 mg of calcium gluconate powder per 1 ml of whole blood.

12. The kit of claim 6, wherein said kaolin powder and calcium gluconate powder are comprised within said coagulation mold.
